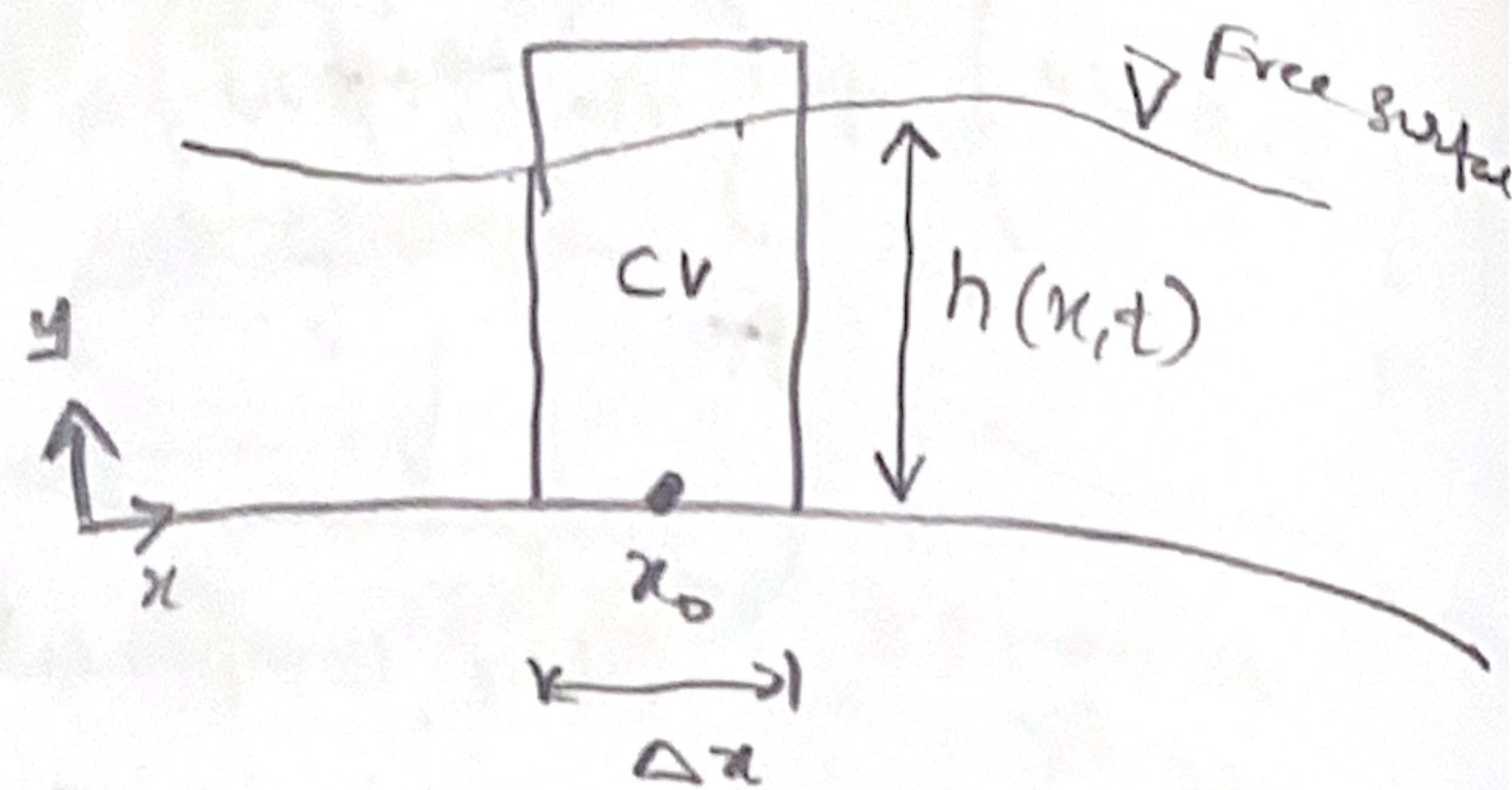


Fluid Mechanics Revision

(1.1) $f(x) = y - h(x, t)$



unit vector normal to

the surface $\hat{n} = \frac{\nabla f}{\|\nabla f\|}$

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \left\langle -\frac{\partial h(x, t)}{\partial x}, 1 \right\rangle$$

$$\|\nabla f\| = \sqrt{1 + \left(\frac{\partial h}{\partial x}\right)^2}$$

$$\hat{n} = \frac{\left(-\frac{\partial h}{\partial x}, 1\right)}{\sqrt{1 + \left(\frac{\partial h}{\partial x}\right)^2}}$$

(1.2) Statement of mass conservation applied to control volume.

$$\text{mass accumulated in CV} = \text{mass inflow} - \text{mass outflow}$$

$$\text{mass of control volume} \Rightarrow \int_{\Delta x} \rho \cdot h(x, t) \cdot dx$$

if fluid is incompressible, ρ is constant

$$M_{CV}(t) = \rho \cdot \int_{\Delta x} h(x, t) \cdot dx$$

Rate of change of mass in control volume

$$\frac{dM(t)}{dt} = \frac{d}{dt} \int_x^{x+\Delta x} \rho \cdot h(x,t) \cdot dx$$

$$\text{mass inflow rate } \dot{m}_{in} = \rho \cdot u(x,y,t) \cdot h(x,t)$$

$$\dot{m}_{out} = \rho \cdot u(x+\Delta x, y, t) \cdot h(x+\Delta x, t)$$

$$\frac{dM(t)}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

$$\Rightarrow \frac{d}{dt} \int_x^{x+\Delta x} \rho \cdot h(x,t) dx = \left[\rho \cdot u(x,y,t) \cdot h(x,t) - \left[\rho \cdot u(x+\Delta x, y, t) \cdot h(x+\Delta x, t) \right] \right]$$

Liebniz Rule

$$\frac{d}{dt} \int_a^b f(x,t) dx = \int_a^b \frac{\partial f(x,t)}{\partial t} dx$$

$$\Rightarrow \int_x^{x+\Delta x} \frac{\partial}{\partial t} (\rho \cdot h(x,t)) \cdot dx = - \left[F(x+\Delta x, t) - F(x, t) \right]$$

$$\left[\text{where } F(x,y,t) = \rho \cdot u(x,y,t) \cdot h(x,t) \right]$$

Divide entire equation by Δx and take $\lim_{\Delta x \rightarrow 0}$

$$\lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \int_x^{x+\Delta x} \frac{\partial}{\partial t} (\rho \cdot h(x,t)) \cdot dx = \lim_{\Delta x \rightarrow 0} \frac{- \left[F(x+\Delta x, y, t) - F(x, y, t) \right]}{\Delta x}$$

$$\Rightarrow \frac{\partial (\rho \cdot h)}{\partial t} = - \frac{\partial F}{\partial x}$$

Definition of the derivative.

Mean value theorem of integrals.

$$\Rightarrow \frac{\partial (s \cdot h)}{\partial t} + \frac{\partial (s \cdot u \cdot h)}{\partial x} = 0$$

For an incompressible flow.

$$-s \cdot \frac{\partial h}{\partial t} + s \cdot \frac{\partial (u \cdot h)}{\partial x} = 0$$

$$\Rightarrow \boxed{\frac{\partial h}{\partial t} + \frac{\partial (u \cdot h)}{\partial x} = 0}$$

{ Accumulation + Divergence flux
= 0.

(1.3) For a steady flow and uniform velocity profile

$$\frac{\partial (h(x,t))}{\partial t} + \frac{\partial (u(x,t) \cdot h)}{\partial x} = 0$$

$u(x, y, t)$ becomes $u(x, t)$

1.4 For a porous surface with percolation at $Q \text{ kg/m}^2/\text{s}$

Mass accumulate = Mass inflow - Mass outflow - loss through
rate rate rate Porous Surface

$$\frac{d}{dt} \int_x^{x+\Delta x} s \cdot h(x,t) \cdot dx = s \cdot u(x,t) \cdot h(x,t) - s \cdot u(x+\Delta x, t) \cdot h(x+\Delta x, t) - Q \cdot \Delta x$$

$$\int_x^{x+\Delta x} \frac{\partial}{\partial t} (s \cdot h(x,t)) \cdot dx = - [F(x+\Delta x, t) - F(x, t)] - Q \cdot \Delta x$$

$$\lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \int_x^{x+\Delta x} \frac{\partial}{\partial t} (s \cdot h(x,t)) \cdot dx = \lim_{\Delta x \rightarrow 0} \frac{- [F(x+\Delta x, t) - F(x, t)]}{\Delta x} - Q$$

⤴

Take it and divide by Δx
D.D.O

$$\frac{\partial (\rho \cdot h)}{\partial t} = - \frac{\partial (\rho u \cdot h)}{\partial x} - Q$$

$$\Rightarrow \frac{\partial (\rho \cdot h)}{\partial t} + \frac{\partial (\rho \cdot u \cdot h)}{\partial x} = -Q$$

$$\Rightarrow \rho \cdot \left[\frac{\partial h}{\partial t} + \frac{\partial (u \cdot h)}{\partial x} \right] = -Q \Rightarrow \boxed{\frac{\partial h}{\partial t} + \frac{\partial (u \cdot h)}{\partial x} = \frac{-Q}{\rho}}$$

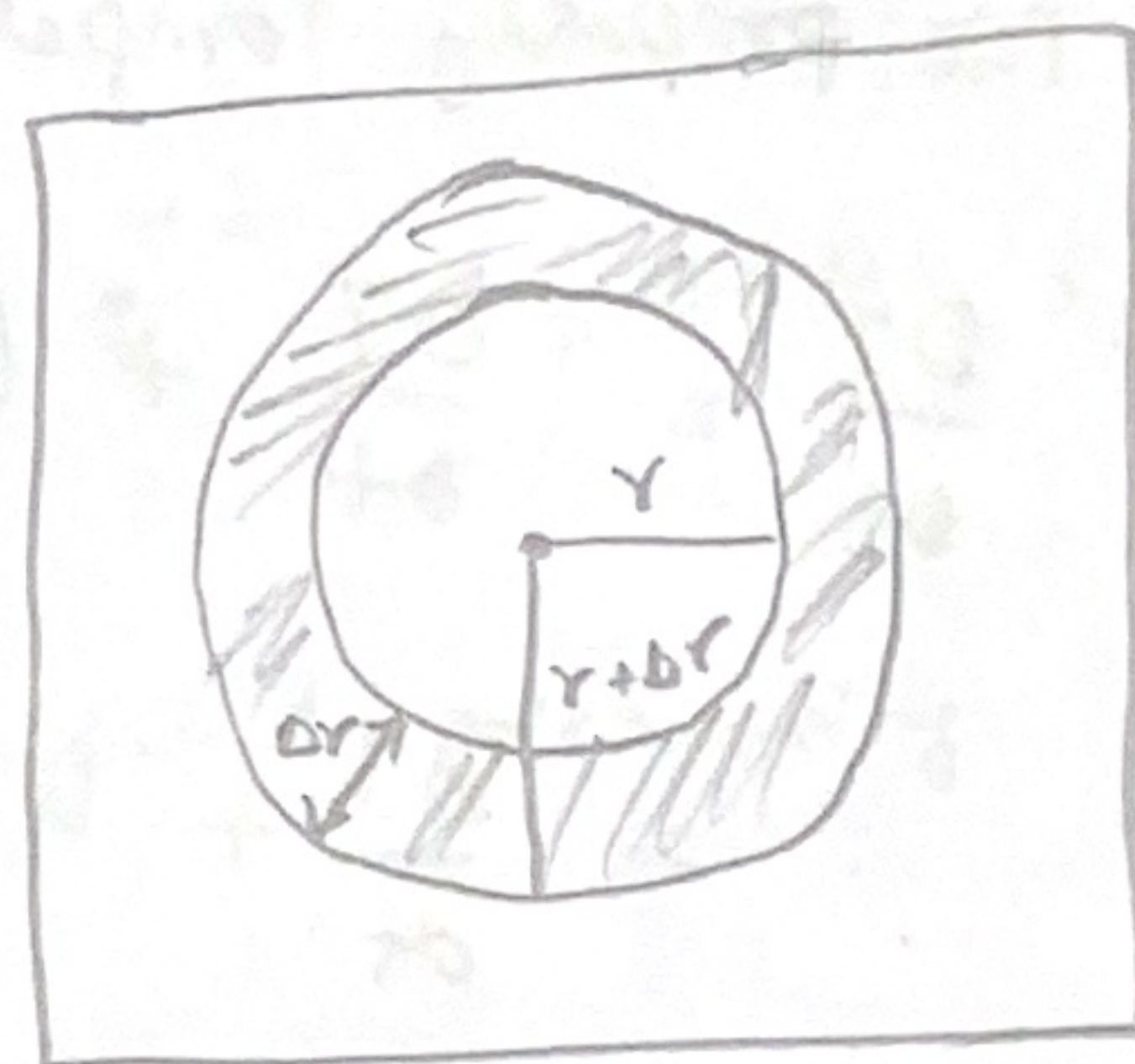
Problem 2

Flow is planar, steady, incompressible and purely radial, u_r is the only non-zero component of velocity.

Mass accumulated in CV = 0

$$\frac{\partial M}{\partial t} = \dot{m}_{in} - \dot{m}_{out} = 0$$

$$\dot{m}_{in} = \rho \cdot u_r(r) \cdot 2\pi r$$



$$\dot{m}_{out} = \rho \cdot u_r(r + \Delta r) \cdot 2\pi (r + \Delta r)$$

$$\rho \cdot u_r(r) \cdot 2\pi r - \rho \cdot u_r(r + \Delta r) \cdot 2\pi (r + \Delta r) = 0$$

$$\Rightarrow r \cdot u_r(r) - (r + \Delta r) \cdot u_r(r + \Delta r) = 0$$

$$\lim_{\Delta r \rightarrow 0} \frac{r \cdot u_r(r) - (r + \Delta r) u_r(r + \Delta r)}{\Delta r} = 0$$

$$\frac{d}{dr} (r \cdot u_r) = 0$$

$$\Rightarrow \boxed{\frac{1}{r} \cdot \frac{d}{dr} (r \cdot u_r) = 0}$$

Divergence form.

Yes vector field represents an incompressible flow.

Problem 3

Substantial derivative is represented by $\frac{D}{Dt}$

It represents the rate of change of a property for moving fluid particle

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \vec{u} \cdot \text{grad } \phi$$

[local change
time change at
fixed point]

[change due to
movement / spatial
gradient]

For property temperature.

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + (u, v, w) \cdot \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$$

$$= \frac{\partial T}{\partial t} + u \cdot \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z}$$

Example A small solid particle travels with velocity of

2 m/s in x-direction through steady temp field.

The temp field increases linearly at rate 0.1 K/m in x-direction.

$$\vec{u} = (2, 0, 0) ; \frac{\partial T}{\partial x} = 0.1 ; \frac{\partial T}{\partial y} = 0 ; \frac{\partial T}{\partial z} = 0.$$

$$\frac{DT}{Dt} = 0 + 0.1 \times 2 + 0 + 0 = 0.2 \text{ K/s.}$$

$$[\text{steady field} \Rightarrow \frac{\partial T}{\partial t} = 0]$$

Problem 4 A fluid particle travels with velocity $(u, v, w) = (5, 5, 0)$ m/s

where $T(x, y, z, t) = 10 \cdot x \cdot e^{-2t}$

$$\frac{\partial T}{\partial t} = \frac{\partial}{\partial t} 10 \cdot x \cdot e^{-2t} = -20 \cdot x \cdot e^{-2t}$$

$$\frac{\partial T}{\partial x} = 10 \cdot e^{-2t} ; \frac{\partial T}{\partial y} = 0 ; \frac{\partial T}{\partial z} = 0$$

$$\frac{DT}{Dt} = -20 \cdot x \cdot e^{-2t} + 5(10 \cdot e^{-2t}) + 0 + 0$$

$$= -20 \cdot x \cdot e^{-2t} + 50 \cdot e^{-2t} = e^{-2t}(50 - 20x)$$

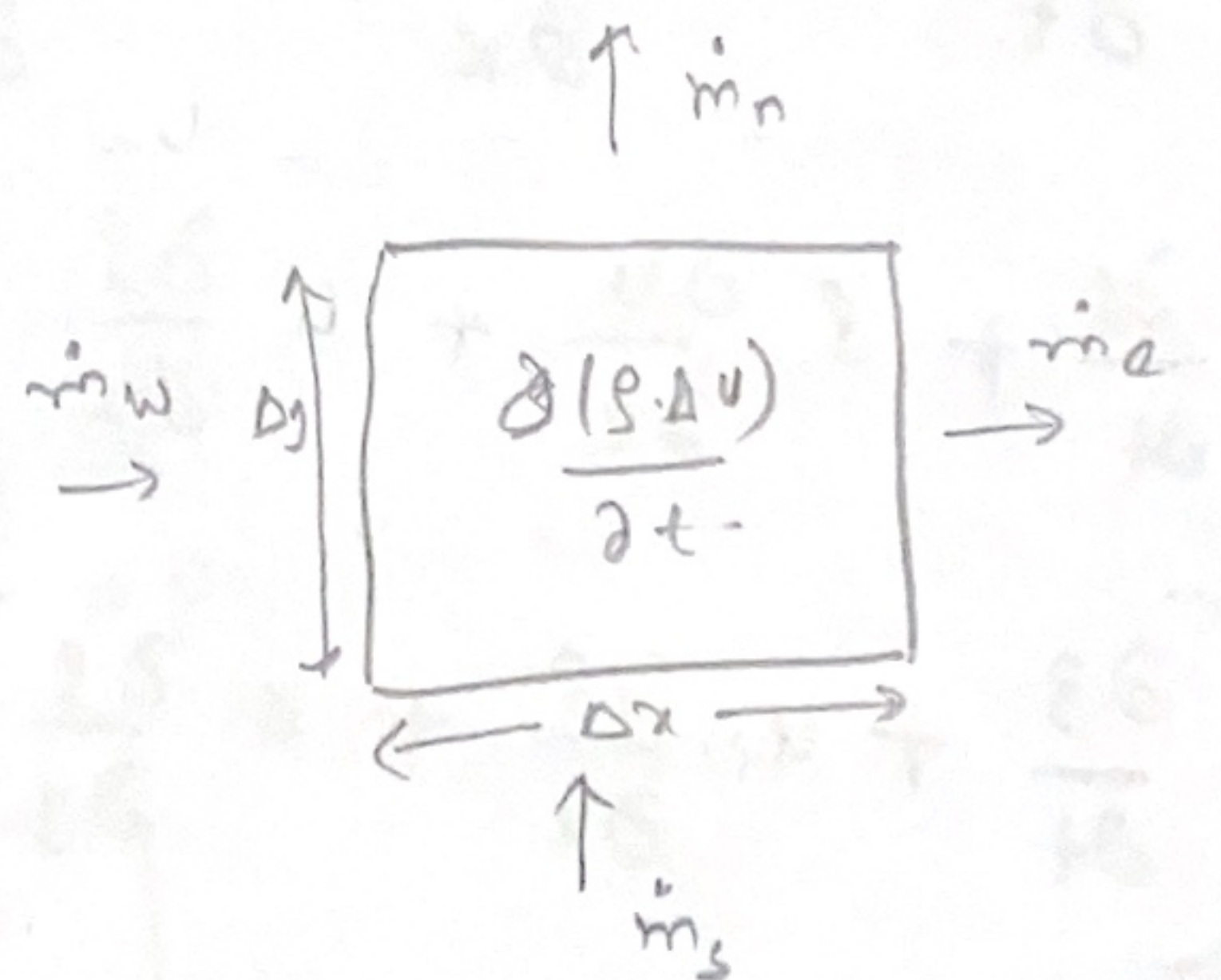
$$\frac{DT}{Dt} = e^{-2t}(50 - 20x)$$

$$\frac{DT(1, 0, 0, 1)}{Dt} = e^{-2 \cdot 1}(50 - 20 \cdot 1) = 30 \cdot e^{-2} = 4.0601, \text{ K/s.}$$

② Derive Continuity equation for any fluid accounting for balance / budget

$$M_{acc} = m_{in} - m_{out}$$

$$\frac{\partial (\rho \cdot \Delta V)}{\partial t} = \dot{m}_w + \dot{m}_s - \dot{m}_e - \dot{m}_n$$



$$\dot{m}_w = (\rho \cdot u)_w \cdot \Delta y$$

$$\dot{m}_s = (\rho \cdot v)_s \cdot \Delta x$$

$$\dot{m}_e = (\rho \cdot u)_e \cdot \Delta y$$

$$\dot{m}_n = (\rho \cdot v)_n \cdot \Delta x$$

$$\Rightarrow \frac{\partial (f \phi)}{\partial t} = (f u)_w \Delta y + (f v)_e \Delta x \\ = (f u)_e \Delta y - (f v)_w \Delta x$$

Adding $\Delta y = \Delta x$ and Δy and div both sides

$$\frac{\partial f}{\partial t} = \frac{(f u)_w - (f u)_e}{\Delta x} + \frac{(f v)_s - (f v)_n}{\Delta y}$$

Limit $\Delta x, \Delta y \rightarrow 0$

$$\frac{\partial f}{\partial t} = \lim_{\Delta x \rightarrow 0} \frac{(f u)_w - (f u)_e}{\Delta x} + \lim_{\Delta y \rightarrow 0} \frac{(f v)_s - (f v)_n}{\Delta y}$$

$$\frac{\partial f}{\partial t} = - \frac{\partial (f u)}{\partial x} + \frac{\partial (f v)}{\partial y}$$

$$\frac{\partial f}{\partial t} + \frac{\partial (f u)}{\partial x} + \frac{\partial (f v)}{\partial y} = 0$$

$$\Rightarrow \frac{\partial f}{\partial t} + f \frac{\partial u}{\partial x} + u \frac{\partial f}{\partial x} + f \frac{\partial v}{\partial y} + v \frac{\partial f}{\partial y} = 0$$

$$\Rightarrow \left[\frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} + v \frac{\partial f}{\partial y} \right] + f \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right] = 0$$

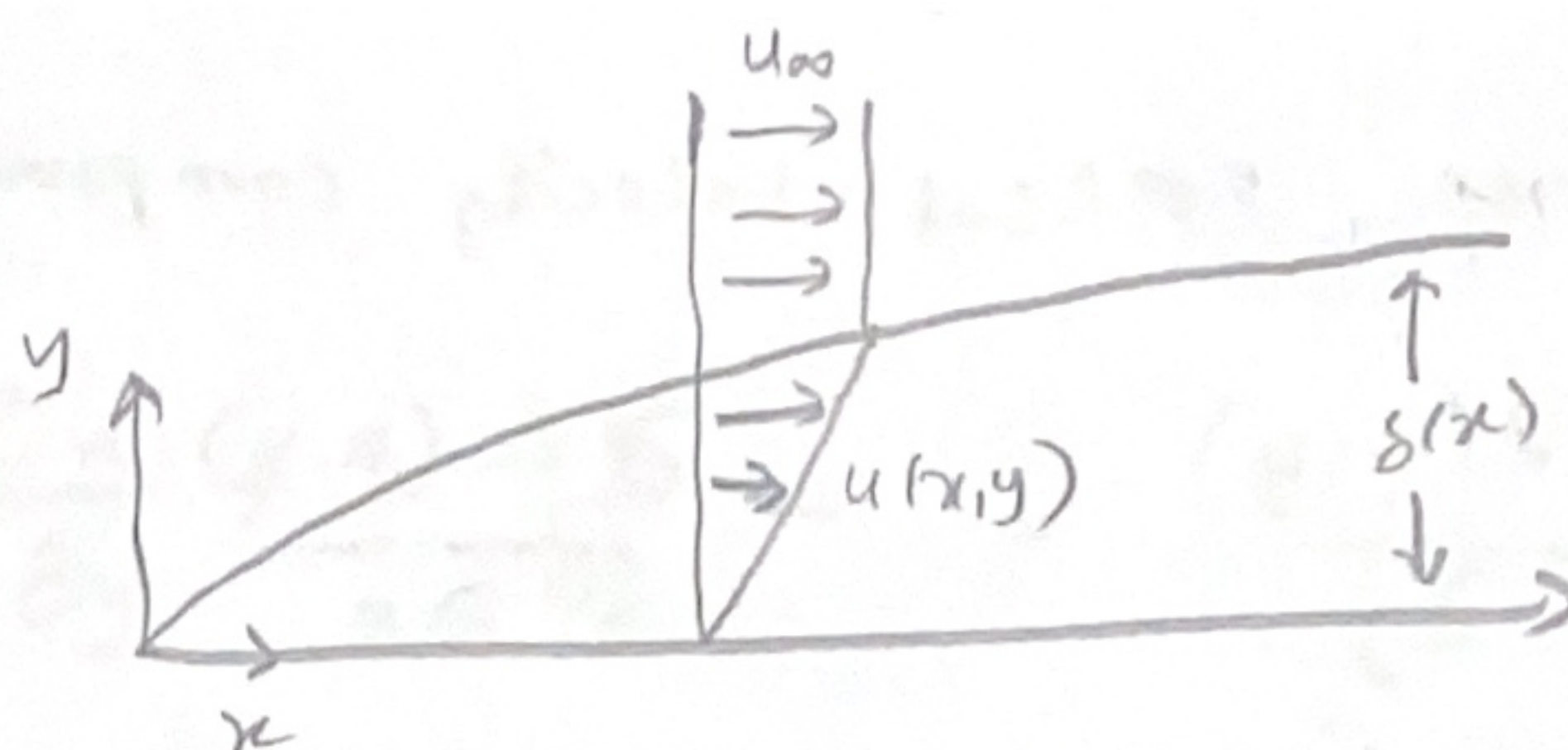
$$\Rightarrow \boxed{\frac{Df}{Dt} + f \left[\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right] = 0}$$

Problem 5

$$u(x,y) = U_{\infty} \frac{y}{\delta}$$

$$\delta = C \cdot \sqrt{x}$$

$$u(x,y) = U_{\infty} \cdot \frac{y}{C \cdot \sqrt{x}}$$



① The flow velocity profile is reasonable because it satisfies the boundary conditions.

at $y = 0$; $u(x,y) = 0$; No slip condition.

at $y = \delta(x)$; $u(x,y) = U_{\infty}$; Boundary layer.

The profile is inversely proportional to $\sqrt{x}$; the flow merges downstream.

② Continuity equation for 2-dim incompressible fluid flow.

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u(x,y))}{\partial x} + \frac{\partial (\rho v(x,y))}{\partial y} = 0.$$

For incompressible flow $\frac{\partial \rho}{\partial t} = 0.$

$$\frac{\partial (\rho \cdot u(x,y))}{\partial x} + \frac{\partial (\rho \cdot v(x,y))}{\partial y} = 0.$$

$$\boxed{\frac{\partial u(x,y)}{\partial x} + \frac{\partial v(x,y)}{\partial y} = 0}$$

③ Reduce vertical velocity component:

$$\frac{\partial v(x,y)}{\partial y} = - \frac{\partial u(x,y)}{\partial x}$$

$$\frac{\partial v(x,y)}{\partial y} = - \frac{\partial}{\partial x} \left(u_{\infty} \cdot \frac{y}{c\sqrt{x}} \right)$$

$$= - \frac{\partial}{\partial x} \left(u_{\infty} \cdot \frac{y \cdot (x)^{-1/2}}{c} \right)$$

$$= - \left(-\frac{1}{2} \right) \cdot \frac{u_{\infty} \cdot y}{c} \cdot x^{-\frac{1}{2}-1}$$

$$= \frac{1}{2} \cdot \frac{u_{\infty} \cdot y}{c} \cdot x^{-3/2}$$

$$\int \frac{\partial v(x,y)}{\partial y} dy = \int \frac{1}{2} \cdot \frac{u_{\infty} \cdot y}{c} \cdot x^{-3/2} dy$$

$$\Rightarrow v(x,y) = \frac{1}{2} \cdot \left(\frac{y^2}{2} \right) \cdot \frac{u_{\infty} \cdot x^{-3/2}}{c} + f(x)$$

$$\Rightarrow v(x,y) = \frac{1}{4c} u_{\infty} \cdot y^2 \cdot x^{-3/2} + f(x)$$

④ applying boundary condition at $y=0$; $v(x,y)=0$

$$f(x) = 0$$

$$v(x,y) = \frac{1}{4c} u_{\infty} \cdot y^2 \cdot x^{-3/2}$$

$$\textcircled{5} \quad \delta(x) = c \cdot \sqrt{x}$$

$$c = \frac{\delta(x)}{\sqrt{x}} = \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1$$

$$\boxed{c = 0.1}$$

⑥ Horizontal acceleration is a_x

$$a_x = \frac{\partial u}{\partial t} + u \cdot \frac{\partial u}{\partial x} + v \cdot \frac{\partial u}{\partial y} + w \cdot \frac{\partial u}{\partial z}$$

$$= \frac{\partial u(x,y)}{\partial t} + u \cdot \frac{\partial u}{\partial x} + v \cdot \frac{\partial u}{\partial y} + w \cdot \frac{\partial u(x,y)}{\partial z}$$

$$a_x = u \cdot \frac{\partial u}{\partial x} + v \cdot \frac{\partial u}{\partial y}$$

$$u(x,y) = \frac{U_\infty \cdot y}{c} \cdot x^{-1/2}$$

$$= \frac{U_\infty \cdot y}{c} \cdot x^{-1/2} \cdot \frac{U_\infty \cdot y}{c} \left(-\frac{1}{2}\right) x^{-3/2} + \frac{1}{4c} \frac{U_\infty^2 y^2}{c} x^{-3/2}$$

$$\frac{U_\infty \cdot y}{c} \cdot x^{-1/2}$$

$$= \left(\frac{U_\infty \cdot y}{c}\right)^2 \left(-\frac{1}{2}\right) x^{-2} + \left(\frac{U_\infty \cdot y}{c}\right)^2 \left(\frac{1}{4}\right) x^{-2}$$

$$= -\frac{1}{4} \left(\frac{U_\infty \cdot y}{c}\right)^2 \cdot \frac{1}{x^2} = -\frac{U_\infty^2 y^2}{4c^2 x^2}$$

⑦ if $x = 1\text{m}$; $y = 0.5\text{ cm}$; $U_\infty = 1\text{ m/s}$
 $= 0.5 \times 10^{-2}\text{ m}$

$$a_x = \frac{-1 \cdot (0.5 \times 10^{-2})^2}{4 \cdot (0.1 \times 10^{-2})^2 \cdot (1)^2} = \frac{-1 \times 0.25}{4 \times 0.01} = \frac{-0.0625}{0.01}$$

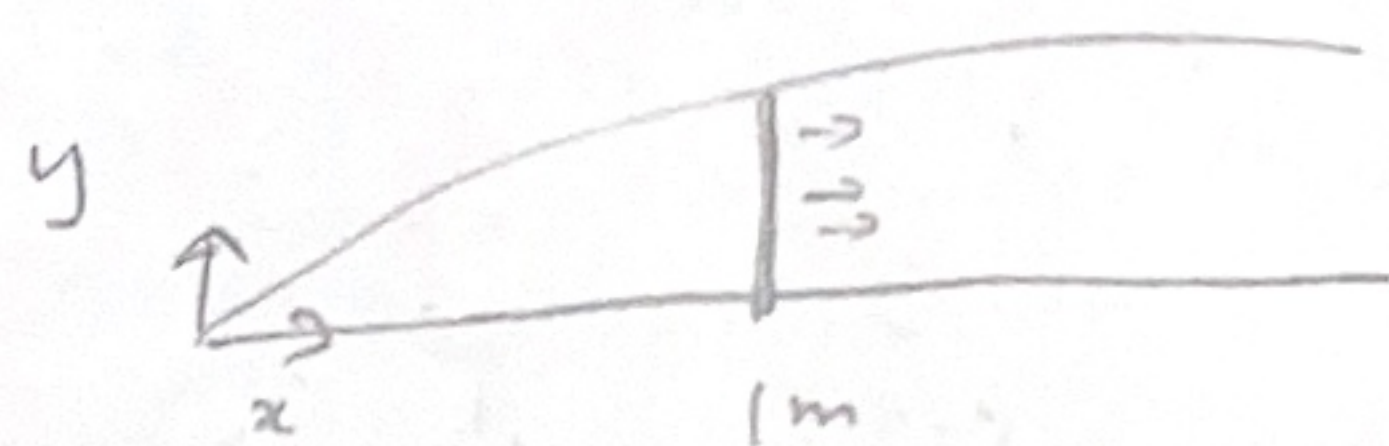
$$= -6.25 \text{ m/s}^2 \quad (\text{decelerating fluid})$$

⑧ local acceleration is experienced at certain time but convective acceleration is experienced as liquid moves through regions of different velocities.

local	convective
Due to time change	due to spatial change
Exists in unsteady flow	exists in steady and unsteady flow
Measured at a fixed point	Experienced by a moving particle
$\frac{\partial u}{\partial t}$	$u \cdot \frac{\partial u}{\partial x} + v \cdot \frac{\partial u}{\partial y}$

⑨ Volumetric flow rate at $x = 1m$.

$$\text{flow rate} = \int_0^{\delta(x)} u(x, y) dy$$



$$= \int_0^{\delta(x)} \frac{U_{\infty} \cdot y}{C \sqrt{x}} \cdot dy$$

$$= \frac{U_{\infty}}{C \sqrt{x}} \int_0^{\delta(x)} y \cdot dy = \frac{U_{\infty}}{\delta(x)} \left[\frac{y^2}{2} \right]_0^{\delta(x)}$$

$$= \frac{U_{\infty}}{\delta(x)} \cdot \frac{\delta(x)^2}{2} = \frac{U_{\infty} \cdot \delta(x)}{2} = \frac{U_{\infty} \cdot C \sqrt{x}}{2}$$

$$U_{\infty} = 1m/s ; C = 0.1 ; x = 1m \quad (\text{assuming constants from before})$$

$$= 100 cm/s ; C = 0.1 ; x = 100cm$$

$$\text{flow rate} = \frac{100 \times 0.1 \times \sqrt{100}}{2} = \frac{100}{2} = 50 \text{ cm}^2/s$$